Introduction to Programming

Lecture No. 1

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Program

"A precise sequence of steps to solve a particular problem"

- A precise sequence of steps to solve a particular problem

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Alan Perlis – Yale University:

“It goes against the grain of modern education to teach children to program. What fun is there in making plans, acquiring discipline in organizing thoughts, devoting attention to detail and learning to be self-critical? “

Critical Skills

- Analysis
- Critical Thinking
- Attention to Detail

- Analysis,critical thinking,attention to detail

Design Recipe

To design a program properly, we must:

- Analyze a problem statement, typically expressed as a word problem
- Express its essence, abstractly and with examples
- Formulate statements and comments in a precise language
- Evaluate and revise the activities in light of checks and tests

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- PAY ATTENTION TO DETAIL
- These skills are useful for anybody
- All assignments in this course should follow the these guidelines

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Computers are

STUPID

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Humans are even more.....

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Think Reuse

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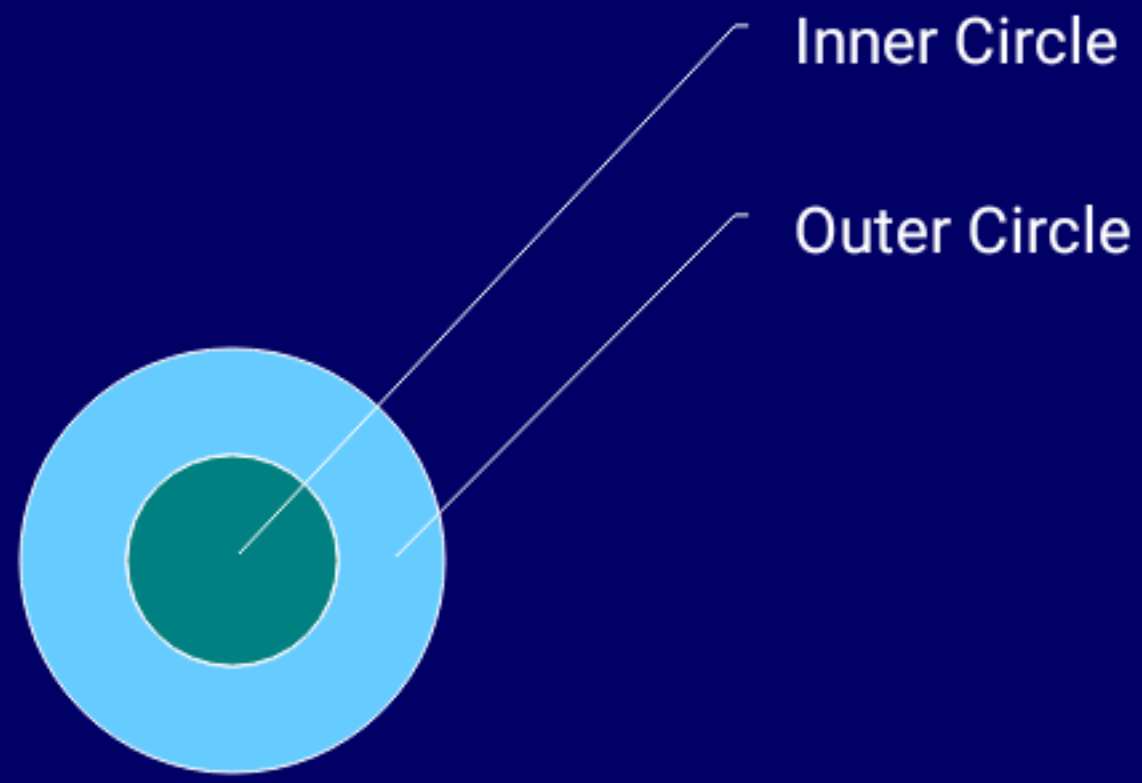
Area of the Ring



Area of Outer Circle — Area of Inner Circle = Area of the Ring

- Think Reuse
- Think User Interface
- Comments liberally

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What is the probability that she gets exactly three letter right i.e. three Letters into their correct envelopes.

Logical Error

Lewis Carol: “Through the Looking Glass”

“Twas brillig, and the slithy toves
Did gyre and gimble in the wabe “

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Course Policy

Policy for the distribution of marks and examination is as follows

- Assignments 15%
- Group discussion 5%
- Midterm 35 %
- Final 45 %

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Books

- Deitel & Deitel :– C++ How to Program
- Kernighan and Ritchie:-
  The C Programming Language

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Course Objectives

Objectives of this course are three fold

- To appreciate the need for a programming language
- To introduce the concept and usability of the structured programming methodology
- To develop proficiency in making useful software using the C language

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Course Contents

To achieve our first two objectives we will be discussing

- Basic Programming constructs and building blocks
- Structured programming
- Structured flowcharts, pseudo-code

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Course Contents

- History of C Language
- Variables and expressions in C
- Control structures and functions
- Arrays and Pointers
- Dynamic memory Allocation

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Course Contents

- File handling
- Structures and Unions
- Flavor of Object oriented programming

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